1.2 The thermodynamic limit

In this section, we will explain how the large numbers of molecules in a typical thermodynamic system mean that it is possible to deal with average quantities. Our explanation proceeds using an analogy: imagine that you are sitting inside a tiny hut with a flat roof. It is raining outside, and you can hear the occasional raindrop striking the roof. The raindrops arrive randomly, so sometimes two arrive close together, but sometimes there is quite a long gap between raindrops. Each raindrop transfers its momentum to the roof and exerts an impulse[2] on it. If you knew the mass and terminal velocity of a raindrop, you could estimate the force on the roof of the hut. The force as a function of time would look like that shown in Fig. 1.1(a), each little blip corresponding to the impulse from one raindrop.

[2] An impulse is the product of force and a time interval. The impulse is equal to the change of momentum.

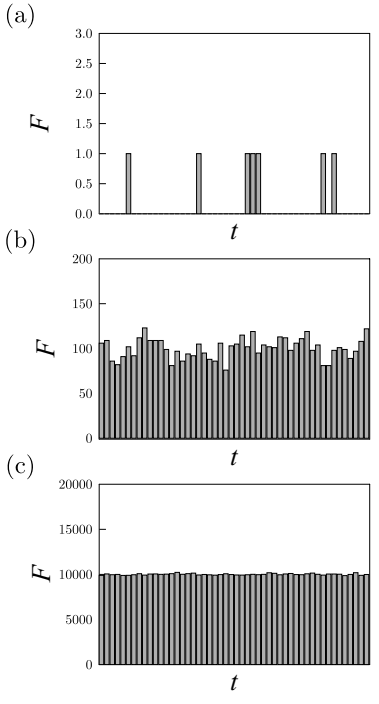


Fig. 1.1 Graphs of the force on a roof as function of time due to falling rain drops.

Now imagine that you are sitting inside a much bigger hut with a flat roof a thousand times the area of the first roof. Many more raindrops will now be falling on the larger roof area and the force as a function of time would look like that shown in Fig. 1.1(b). Now scale up the area of the flat roof by a further factor of one hundred and the force would look like that shown in Fig. 1.1(c). Notice two key things about these graphs:

(1) The force, on average, gets bigger as the area of the roof gets bigger. This is not surprising because a bigger roof catches more raindrops.

(2) The *fluctuations* in the force get smoothed out and the force looks like it stays much closer to its average value. In fact, the fluctuations are still big but, as the area of the roof increases, they grow more slowly than the average force does.

The force grows with area, so it is useful to consider the **pressure** which is defined as

$$\text{pressure} = \frac{\text{force}}{\text{area}}. \tag{1.3}$$

The average pressure due to the falling raindrops will not change as the area of the roof increases, but the fluctuations in the pressure will decrease. In fact, we can completely ignore the fluctuations in the pressure in the limit that the area of the roof *grows to infinity*. This is precisely analogous to the limit we refer to as the **thermodynamic limit**.

Consider now the molecules of a gas which are bouncing around in a container. Each time the molecules bounce off the walls of the container, they exert an impulse on the walls. The net effect of all these impulses is a pressure, a force per unit area, exerted on the walls of the container. If the container were very small, we would have to worry about fluctuations in the pressure (the random arrival of individual molecules on the wall, much like the raindrops in Fig. 1.1(a)). However, in most cases that one meets, the number of molecules in a container of gas is extremely large, so these fluctuations can be ignored and the pressure of the gas appears to be completely uniform. Again, our description of the pressure of this

system can be said to be 'in the thermodynamic limit', where we have let the number of molecules be regarded as tending to infinity in such a way that the density of the gas is a constant.

Suppose that the container of gas has volume V, that the temperature is T, the pressure is p and the kinetic energy of all the gas molecules adds up to U. Imagine slicing the container of gas in half with an imaginary plane, and now just focus your attention on the gas on one side of the plane. The volume of this half of the gas, let's call it V^*, is by definition half that of the original container, i.e.

$$V^* = \frac{V}{2}. \tag{1.4}$$

The kinetic energy of this half of the gas, let's call it U^*, is clearly half that of the total kinetic energy, i.e.

$$U^* = \frac{U}{2}. \tag{1.5}$$

However, the pressure p^* and the temperature T^* of this half of the gas are the same as for the whole container of gas, so that

$$p^* = p \tag{1.6}$$
$$T^* = T. \tag{1.7}$$

Variables which scale with the system size, like V and U, are called **extensive variables**. Those which are independent of system size, like p and T, are called **intensive variables**.

Thermal physics evolved in various stages and has left us with various approaches to the subject:

- The subject of **classical thermodynamics** deals with macroscopic properties, such as pressure, volume and temperature, without worrying about the underlying microscopic physics. It applies to systems which are sufficiently large that microscopic fluctuations can be ignored, and it does not assume that there is an underlying atomic structure to matter.
- The **kinetic theory of gases** tries to determine the properties of gases by considering probability distributions associated with the motions of individual molecules. This was initially somewhat controversial since the existence of atoms and molecules was doubted by many until the late nineteenth and early twentieth centuries.
- The realization that atoms and molecules exist led to the development of **statistical mechanics**. Rather than starting with descriptions of macroscopic properties (as in thermodynamics) this approach begins with trying to describe the individual microscopic states of a system and then uses statistical methods to derive the macroscopic properties from them. This approach received an additional impetus with the development of **quantum theory** which showed explicitly how to describe the microscopic quantum states

of different systems. The thermodynamic behavior of a system is then asymptotically approximated by the results of statistical mechanics in the *thermodynamic limit*, i.e. as the number of particles tends to infinity (with intensive quantities such as pressure and density remaining finite).

In the next section, we will state the *ideal gas law* which was first found experimentally but can be deduced from the kinetic theory of gases (see Chapter 6).

1.3 The ideal gas

Experiments on gases show that the pressure p of a volume V of gas depends on its temperature T. For example, a fixed amount of gas at constant temperature obeys

$$p \propto 1/V, \tag{1.8}$$

a result which is known as **Boyle's law** (sometimes as the Boyle–Mariotte law); it was discovered experimentally by Robert Boyle (1627–1691) in 1662 and independently by Edmé Mariotte (1620–1684) in 1676. At constant pressure, the gas also obeys

$$V \propto T, \tag{1.9}$$

where T is measured in Kelvin. This is known as **Charles' law** and was discovered experimentally, in a crude fashion, by Jacques Charles (1746–1823) in 1787, and more completely by Joseph Louis Gay-Lussac (1778–1850) in 1802, though their work was partly anticipated by Guillaume Amontons (1663–1705) in 1699, who also noticed that a fixed volume of gas obeys

$$p \propto T, \tag{1.10}$$

a result that Gay-Lussac himself found independently in 1809 and is often known as **Gay-Lussac's law**.[3]

These three empirical laws can be combined to give

$$pV \propto T. \tag{1.11}$$

It turns out that, if there are N molecules in the gas, this finding can be expressed as follows:

$$\boxed{pV = Nk_{\mathrm{B}}T.} \tag{1.12}$$

This is known as the **ideal gas equation**, and the constant k_{B} is known as the **Boltzmann constant**.[4] We now make some comments about the ideal gas equation.

- We have stated this law purely as an empirical law, observed in experiment. We will derive it from first principles using the kinetic theory of gases in Chapter 6. This theory assumes that a gas can be modelled as a collection of individual tiny particles which can bounce off the walls of the container, and each other (see Fig. 1.2).

[3] Note that none of these scientists expressed temperature in this way, since the Kelvin scale and absolute zero had yet to be invented. For example, Gay-Lussac found merely that $V = V_0(1 + \alpha\tilde{T})$, where V_0 and α are constants and $\tilde{T}$ is temperature in his scale.

[4] It takes the numerical value $k_{\mathrm{B}} = 1.3807\times10^{-23}\,\mathrm{J\,K^{-1}}$. We will meet this constant again in eqn 4.7.

- Why do we call it 'ideal'? The microscopic justification which we will present in Chapter 6 proceeds under various assumptions: (i) we assume that there are no intermolecular forces, so that the molecules are not attracted to each other; (ii) we assume that molecules are point-like and have zero size. These are idealized assumptions and so we do not expect the ideal gas model to describe real gases under all circumstances. However, it does have the virtue of simplicity: eqn 1.12 is simple to write down and remember. Perhaps more importantly, it does describe gases quite well under quite a wide range of conditions.
- The ideal gas equation forms the basis of much of our study of classical thermodynamics. Gases are common in nature: they are encountered in astrophysics and atmospheric physics and it is gases which are used to drive engines, and thermodynamics was invented to try and understand engines. Therefore this equation is fundamental in our treatment of thermodynamics and should be memorized.
- The ideal gas law, however, doesn't describe all important gases, and several chapters in this book are devoted to seeing what happens when various assumptions fail. For example, the ideal gas equation assumes that the gas molecules move non-relativistically. When this is not the case, we have to develop a model of relativistic gases (see Chapter 25). At low temperatures and high densities, gas molecules do attract one another (this must occur for liquids and solids to form) and this is considered in Chapters 26, 27 and 28. Furthermore, when quantum effects are important we need a model of quantum gases, and this is outlined in Chapter 30.
- Of course, thermodynamics applies also to systems which are not gaseous (so the ideal gas equation, though useful, is not a cure for all ills), and we will look at the thermodynamics of rods, bubbles and magnets in Chapter 17.

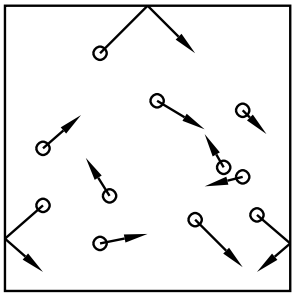


Fig. 1.2 In the kinetic theory of gases, a gas is modelled as a number of individual tiny particles which can bounce off the walls of the container, and each other.

1.4 Combinatorial problems

Even larger numbers than N_A occur in problems involving combinations, and these turn out to be very important in thermal physics. The following example illustrates a simple combinatorial problem which captures the essence of what we are going to have to deal with.

Example 1.3

Let us imagine that a certain system contains ten atoms. Each of these atoms can exist in one of two states, according to whether it has zero units or one unit of energy. These 'units' of energy are called **quanta** of energy. How many distinct arrangements of quanta are possible for this system if you have at your disposal (a) 10 quanta of energy; (b) 4 quanta of energy?



Fig. 1.3 Ten atoms which can accommodate four quanta of energy. An atom with a single quantum of energy is shown as a filled circle, otherwise it is shown as an empty circle. One configuration is shown here.

Solution:
We can represent the ten atoms by drawing ten boxes; an empty box signifies an atom with zero quanta of energy; a filled box signifies an atom with one quantum of energy (see Fig. 1.3). We give two methods for calculating the number of ways of arranging r quanta among n atoms:

(1) In the first method, we realize that the first quantum can be assigned to any of the n atoms, the second quantum can be assigned to any of the remaining atoms (there are $n-1$ of them), and so on until the rth quantum can be assigned to any of the remaining $n-r+1$ atoms. Thus our first guess for the number of possible arrangements of the r quanta we have assigned, is $\Omega_{\text{guess}} = n \times (n-1) \times (n-2) \times \ldots \times (n-r+1)$. This can be simplified as follows:

$$\Omega_{\text{guess}} = \frac{n \times (n-1) \times (n-2) \times \ldots \times 1}{(n-r) \times (n-r-1) \times \ldots \times 1} = \frac{n!}{(n-r)!}. \qquad (1.13)$$

However, this assumes that we have labelled the quanta as 'the first quantum', 'the second quantum' etc. In fact, we don't care which quantum is which because they are indistinguishable. We can rearrange the r quanta in any one of $r!$ arrangements. Hence our answer Ω_{guess} needs to be divided by $r!$, so that the number Ω of unique arrangements is

$$\Omega = \frac{n!}{(n-r)!\,r!} \equiv {}^{n}C_{r}, \qquad (1.14)$$

where ${}^{n}C_{r}$ is the symbol for a **combination**.[5]

(2) In the second method, we recognize that there are r atoms each with one quantum and $n-r$ atoms with zero quanta. The number of arrangements is then simply the number of ways of arranging r ones and $n-r$ zeros. There are $n!$ ways of arranging a sequence of n distinguishable symbols. If r of these symbols are the same (all ones), there are $r!$ ways of arranging these without changing the pattern. If the remaining $n-r$ symbols are all the same (all zeros), there are $(n-r)!$ ways of arranging these without changing the pattern. Hence we again find that

$$\Omega = \frac{n!}{(n-r)!\,r!}. \qquad (1.15)$$

[5]Other symbols sometimes used for ${}^{n}C_{r}$ include ${}^{n}_{r}C$ and $\binom{n}{r}$.

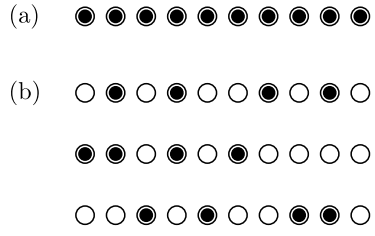


Fig. 1.4 Each row shows the ten atoms which can accommodate r quanta of energy. An atom with a single quantum of energy is shown as a filled circle, otherwise it is shown as an empty circle. (a) For $r = 10$ there is only one possible configuration. (b) For $r = 4$, there are 210 possibilities, of which three are shown.

For the specific cases shown in Fig. 1.4:

(a) $n = 10$, $r = 10$, so $\Omega = 10!/(10! \times 0!) = 1$. This one possibility, with each atom having a quantum of energy, is shown in Fig. 1.4(a).

(b) $n = 10$, $r = 4$, so $\Omega = 10!/(6! \times 4!) = 210$. A few of these possibilities are shown in Fig. 1.4(b).

If instead we had chosen 10 times as many atoms (so $n = 100$) and 10 times as many quanta, the numbers for (b) would have come out much much bigger. In this case, we would have $r = 40$, $\Omega \sim 10^{28}$. A further factor of 10 sends these numbers up much further, so for $n = 1000$ and $r = 400$, $\Omega \sim 10^{290}$ – a staggeringly large number.

The numbers in the above example are so large because factorials increase very quickly. In our example we treated 10 atoms; we are clearly going to run into trouble when we are going to deal with a mole of atoms, i.e. when $n = 6 \times 10^{23}$.

One way of bringing large numbers down to size is to look at their logarithms.[6] Thus, if Ω is given by eqn 1.15, we could calculate

$$\ln \Omega = \ln(n!) - \ln((n-r)!) - \ln(r!). \tag{1.16}$$

[6] We will use 'ln' to signify log to the base e, i.e. $\ln = \log_e$. This is known as the natural logarithm.

This expression involves the logarithm of a factorial, and it is going to be very useful to be able to evaluate this. Most pocket calculators have difficulty in evaluating factorials above 69! (because $70! > 10^{100}$ and many pocket calculators give an overflow error for numbers above 9.999×10^{99}), so some low cunning will be needed to overcome this. Such low cunning is provided by an expression termed **Stirling's formula**:

$$\boxed{\ln n! \approx n \ln n - n.} \tag{1.17}$$

This expression[7] is derived in Appendix C.3.

[7] As shown in Appendix C.3, it is slightly more accurate to use the formula $\ln n! \approx n \ln n - n + \frac{1}{2} \ln 2\pi n$, but this only gives a significant advantage when n is not too large.

Example 1.4

Estimate the order of magnitude of $10^{23}!$.
Solution:
Using Stirling's formula, we can estimate

$$\ln 10^{23}! \approx 10^{23} \ln 10^{23} - 10^{23} = 5.2 \times 10^{24}, \tag{1.18}$$

and hence

$$10^{23}! = \exp(\ln 10^{23}!) \approx \exp(5.20 \times 10^{24}). \tag{1.19}$$

We have our answer in the form e^x, but we would really like it as ten to some power. Now if $e^x = 10^y$, then $y = x/\ln 10$ and hence

$$10^{23}! \approx 10^{2.26 \times 10^{24}}. \tag{1.20}$$

Just pause for a moment to take in how big this number is. It is roughly one followed by about 2.26×10^{24} zeros! Our claim that combinatorial numbers are big seems to be justified!

1.5 Plan of the book

This book aims to introduce the concepts of thermal physics one by one, steadily building up the techniques and ideas which make up the subject. Part I contains various preliminary topics. In Chapter 2 we define heat and introduce the idea of heat capacity. In Chapter 3, the ideas of probability are presented for discrete and continuous distributions. (For

Heat

2

In this Chapter, we will introduce the concepts of heat and heat capacity.

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2.1 A definition of heat

We all have an intuitive notion of what heat is: sitting next to a roaring fire in winter, we *feel* its heat warming us up, increasing our temperature; lying outside in the sunshine on a warm day, we *feel* the Sun's heat warming us up. In contrast, holding a snowball, we feel heat leaving our hand and transferring to the snowball, making our hand feel cold. Heat seems to be some sort of energy transferred from hot things to cold things when they come into contact. We therefore make the following definition:

heat is *energy in transit*.

We now stress a couple of important points about this definition.

(1) Experiments suggest that heat spontaneously transfers from a hotter body to a colder body when they are in contact, and not in the reverse direction. However, there are circumstances when it is possible for heat to go in the reverse direction. A good example of this is a kitchen freezer: you place food, initially at room temperature, into the freezer and shut the door; the freezer then sucks heat out of the food and cools the food down to below freezing point. Heat is being transferred from your warmer food to the colder freezer, apparently in the 'wrong' direction. Of course, to achieve this, you have to be paying your electricity bill and therefore be putting in energy to your freezer. If there is a power cut, heat will slowly leak back into the freezer from the warmer kitchen and thaw out all your frozen food. This shows that it is possible to reverse the direction of heat flow, but only if you intervene by putting additional energy in. We will return to this point in Section 13.5 when we consider refrigerators, but for now let us note that we are defining heat as *energy in transit* and not hard-wiring into the definition anything about which direction it goes.

(2) The 'in transit' part of our definition is very important. Though you can add heat to an object, you *cannot* say that 'an object contains a certain quantity of heat.' This is very different to the case of the fuel in your car: you can add fuel to your car, and you

are quite entitled to say that your car 'contains a certain quantity of fuel'. You even have a gauge for measuring it! But heat is quite different. Objects do not and cannot have gauges which read out how much heat they contain, because heat only makes sense when it is 'in transit'.[1]

To see this, consider your cold hands on a chilly winter day. You can increase the temperature of your hands in two different ways: (i) by adding heat, for example by putting your hands close to something hot, like a roaring fire; (ii) by rubbing your hands together. In one case you have added heat from the outside, in the other case you have not added any heat but have done some work. In both cases, you end up with the same final situation: hands which have increased in temperature. There is no physical difference between hands which have been warmed by heat and hands which have been warmed by work.[2]

[1] We will see later that objects can contain a certain quantity of *energy*, so it is possible, at least in principle, to have a gauge which reads out how much energy is contained.

[2] We have made this point by giving a plausible example, but in Chapter 11 we will show using more mathematical arguments that heat only makes sense as energy 'in transit'.

Heat is measured in joules (J). The rate of heating has the units of watts (W), where $1\,\mathrm{W}{=}1\,\mathrm{J\,s^{-1}}$.

Example 2.1

A 1 kW electric heater is switched on for ten minutes. How much heat does it produce?

Solution:

Ten minutes equals 600 s, so the heat Q is given by

$$Q = 1\,\mathrm{kW} \times 600\,\mathrm{s} = 600\,\mathrm{kJ}. \tag{2.1}$$

Notice in this last example that the power in the heater is supplied by electrical work. Thus it is possible to produce heat by doing work. We will return to the question of whether one can produce work from heat in Chapter 13.

2.2 Heat capacity

In the previous section, we explained that it is not possible for an object to contain a certain quantity of heat, because heat is defined as 'energy in transit'. It is therefore with a somewhat heavy heart that we turn to the topic of 'heat capacity', since we have argued that objects have no capacity for heat! (This is one of those occasions in physics when decades of use of a name have made it completely standard, even though it is really the wrong name to use.) What we are going to derive in this section might be better termed 'energy capacity', but to do this would put us at odds with common usage throughout physics. All of this being said, we can proceed quite legitimately by asking the following simple question:

How much heat needs to be supplied to an object to raise its temperature by a small amount $\mathrm{d}T$*?*

The answer to this question is the heat $\mathrm{d}Q = C\,\mathrm{d}T$, where we define the **heat capacity** C of an object using

$$C = \frac{\mathrm{d}Q}{\mathrm{d}T}. \tag{2.2}$$

As long as we remember that heat capacity tells us simply how much heat is needed to warm an object (and is nothing about the capacity of an object for heat) we shall be on safe ground. As can be inferred from eqn 2.2, the heat capacity C has units $\mathrm{J\,K^{-1}}$.

As shown in the following example, although objects have a heat capacity, one can also express the heat capacity of a particular substance *per unit mass*, or *per unit volume*.[3]

[3] We will use the symbol C to represent a heat capacity, whether of an object, or per unit volume, or per mole. We will always state which is being used. The heat capacity per unit mass is distinguished by the use of the lower-case symbol c. We will usually reserve the use of subscripts on the heat capacity to denote the constraint being applied (see later).

Example 2.2

The heat capacity of 0.125 kg of water is measured to be $523\,\mathrm{J\,K^{-1}}$ at room temperature. Hence calculate the heat capacity of water (a) per unit mass and (b) per unit volume.

Solution:

(a) The heat capacity per unit mass c is given by dividing the heat capacity by the mass, and hence

$$c = \frac{523\,\mathrm{J\,K^{-1}}}{0.125\,\mathrm{kg}} = 4.184 \times 10^3\,\mathrm{J\,K^{-1}\,kg^{-1}}\ . \tag{2.3}$$

(b) The heat capacity per unit volume C is obtained by multiplying the previous answer by the density of water, namely $1000\,\mathrm{kg\,m^{-3}}$, so that

$$C = 4.184 \times 10^3\,\mathrm{J\,K^{-1}\,kg^{-1}} \times 1000\,\mathrm{kg\,m^{-3}} = 4.184 \times 10^6\,\mathrm{J\,K^{-1}\,m^{-3}}. \tag{2.4}$$

The heat capacity per unit mass c occurs quite frequently, and it is given a special name: the **specific heat capacity**.

Example 2.3

Calculate the specific heat capacity of water.

Solution:

This is given in answer (a) from the previous example: the specific heat capacity of water is $4.184 \times 10^3\,\mathrm{J\,K^{-1}\,kg^{-1}}$.

Also useful is the **molar heat capacity**, which is the heat capacity of one mole of the substance.

Example 2.4

Calculate the molar heat capacity of water. (The molar mass of water is 18 g.)

Solution:

The molar heat capacity is obtained by multiplying the specific heat capacity by the molar mass, and hence

$$C = 4.184 \times 10^3 \,\mathrm{J\,K^{-1}\,kg^{-1}} \times 0.018 \,\mathrm{kg} = 75.2 \,\mathrm{J\,K^{-1}\,mol^{-1}}. \qquad (2.5)$$

When we think about the heat capacity of a gas, there is a further complication.[4] We are trying to ask the question: how much heat should you add to raise the temperature of our gas by one degree Kelvin? But we can imagine doing the experiment in two ways (see also Fig. 2.1):

(1) Place our gas in a sealed box and add heat (Fig. 2.1(a)). As the temperature rises, the gas will not be allowed to expand because its volume is fixed, so its pressure will increase. This method is known as heating *at constant volume*.

(2) Place our gas in a chamber connected to a piston and heat it (Fig. 2.1(b)). The piston is well lubricated, and so will slide in and out to maintain the pressure in the chamber to be identical to that in the lab. As the temperature rises, the piston is forced out (doing work against the atmosphere) and the gas is allowed to expand, keeping its pressure constant. This method is known as heating *at constant pressure*.

[4]This complication is there for liquids and solids, but doesn't make such a big difference.

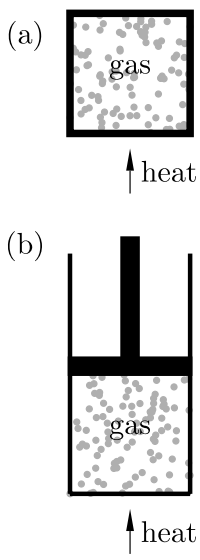


Fig. 2.1 Two methods of heating a gas: (a) constant volume, (b) constant pressure.

In both cases, we are applying a **constraint** to the system, either constraining the volume of the gas to be fixed, or constraining the pressure of the gas to be fixed. We need to modify our definition of heat capacity given in eqn 2.2, and hence we define two new quantities: C_V is the heat capacity *at constant volume* and C_p is the heat capacity *at constant pressure*. We can write them using partial differentials as follows:

$$C_V = \left(\frac{\partial Q}{\partial T}\right)_V, \qquad (2.6)$$

$$C_p = \left(\frac{\partial Q}{\partial T}\right)_p. \qquad (2.7)$$

We expect that C_p will be bigger than C_V for the simple reason that more heat will need to be added when heating at constant pressure than when heating at constant volume. This is because in the latter case additional energy will be expended on doing work on the atmosphere as the gas expands. It turns out that indeed C_p is bigger than C_V in practice.[5]

[5]We will *calculate* the relative sizes of C_V and C_p in Section 11.3.

Example 2.5

The specific heat capacity of helium gas is measured to be $3.12\,\mathrm{kJ\,K^{-1}\,kg^{-1}}$ at constant volume and $5.19\,\mathrm{kJ\,K^{-1}\,kg^{-1}}$ at constant pressure. Calculate the molar heat capacities. (The molar mass of helium is 4 g.)

Solution:

The molar heat capacity is obtained by multiplying the specific heat capacity by the molar mass, and hence

$$C_V = 12.48\,\mathrm{J\,K^{-1}\,mol^{-1}}, \tag{2.8}$$

$$C_p = 20.76\,\mathrm{J\,K^{-1}\,mol^{-1}}. \tag{2.9}$$

(Interestingly, these answers are almost exactly $\frac{3}{2}R$ and $\frac{5}{2}R$. We will see why in Section 11.3.)

Chapter summary

- In this chapter, the concepts of heat and heat capacity have been introduced.
- Heat is 'energy in transit'.
- The heat capacity C of an object is given by $C = \mathrm{d}Q/\mathrm{d}T$. The heat capacity of a substance can also be expressed per unit volume or per unit mass (in the latter case it is called *specific heat capacity*).

Exercises

(2.1) Using data from this chapter, estimate the energy needed to (a) boil enough tap water to make a cup of tea, (b) heat the water for a bath.

(2.2) The world's oceans contain approximately 10^{21} kg of water. Estimate the total heat capacity of the world's oceans.

(2.3) The world's power consumption is currently about 13 TW, and growing! (1 TW= 10^{12} W.) Burning one ton of crude oil (which is nearly seven barrels worth) produces about 42 GJ (1 GJ= 10^9 J). If the world's total power needs were to come from burning oil (a large fraction currently does), how much oil would we be burning per second?

(2.4) The molar heat capacity of gold is $25.4\,\mathrm{J\,mol^{-1}\,K^{-1}}$. Its density is $19.3\times10^3\,\mathrm{kg\,m^{-3}}$. Calculate the specific heat capacity of gold and the heat capacity per unit volume. What is the heat capacity of 4×10^6 kg of gold? (This is roughly the holdings of Fort Knox.)

(2.5) Two bodies, with heat capacities C_1 and C_2 (assumed independent of temperature) and initial temperatures T_1 and T_2 respectively, are placed in thermal contact. Show that their final temperature T_f is given by $T_\mathrm{f} = (C_1T_1 + C_2T_2)/(C_1 + C_2)$. If C_1 is much larger than C_2, show that $T_\mathrm{f} \approx T_1 + C_2(T_2 - T_1)/C_1$.

3 Probability

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Life is full of uncertainties, and has to be lived according to our best guesses based on the information available to us. This is because the chain of events that lead to various outcomes can be so complex that the exact outcomes are unpredictable. Nevertheless, things can still be said even in an uncertain world: for example, it is more helpful to know that there is a 20% chance of rain tomorrow than that the weather forecaster has absolutely no idea; or worse still that he/she claims that there will definitely be no rain, when there might be! Probability is therefore an enormously useful and powerful subject, since it can be used to *quantify* uncertainty.

The foundations of probability theory were laid by the French mathematicians Pierre de Fermat (1601–1665) and Blaise Pascal (1623–1662), in a correspondence in 1654 which originated from a problem set to them by a gentleman gambler. The ideas proved to be intellectually infectious and the first probability textbook was written by the Dutch physicist Christian Huygens (1629–1695) in 1657, who applied it to the working out of life expectancy. Probability was thought to be useful only for determining possible outcomes in situations in which we lacked complete knowledge. The supposition was that if we could know the motions of all particles at the microscopic level, we could determine every outcome precisely. In the twentieth century, the discovery of quantum theory has led to the understanding that, at the microscopic level, outcomes are purely probabilistic.

Probability has had a huge impact on thermal physics. This is because we are often interested in systems containing huge numbers of particles, so that predictions based on probability turn out to be precise enough for most purposes. In a thermal physics problem, one is often interested in the values of quantities which are the sum of many small contributions from individual atoms. Though each atom behaves differently, the *average* behaviour is what comes through, and therefore it becomes necessary to be able to extract average values from probability distributions.

In this chapter, we will define some basic concepts in probability theory. Let us begin by stating that the probability of occurrence of a particular event, taken from a finite set of possible events, is zero if that event is impossible, is one if that event is certain, and takes a value somewhere in between zero and one if that event is possible but not certain. We begin by considering two different types of probability distribution: *discrete* and *continuous*.

4 Temperature and the Boltzmann factor

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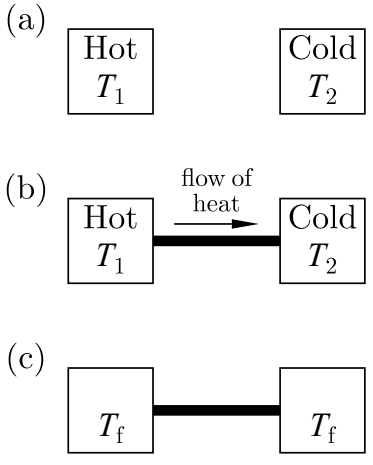


Fig. 4.1 (a) Two objects at different temperatures. (b) The objects are now placed in thermal contact and heat flows from the hot object to the cold object. (c) After a long time, the two objects have the same final temperature T_f.

In this chapter, we will explore the concept of *temperature* and show how it can be defined in a statistical manner. This leads to the idea of a *Boltzmann distribution* and a *Boltzmann factor*. Now of course the concept of **temperature** seems such an intuitively obvious one that you might wonder why we need a whole chapter to discuss it. Temperature is simply a measure of 'hotness' or 'coldness', so that we say that a hot body has a higher temperature than a cold one. For example, as shown in Fig. 4.1(a) if an object has temperature T_1 and is hotter than a second body with temperature T_2, we expect that $T_1 > T_2$. But what do these numbers T_1 and T_2 signify? What does temperature actually mean?

4.1 Thermal equilibrium

To begin to answer these questions, let us consider what happens if our hot and cold bodies are placed in **thermal contact** which means that they are able to exchange energy. As described in Chapter 2, heat is 'energy in transit' and experiment suggests that, if nothing else is going on,[1] heat will always flow from the hotter body to the colder body, as shown in Fig. 4.1(b). This is backed up by our experience of the world: we always seem to burn ourselves when we touch something very hot (heat flows into us from the hot object) and become very chilled when we touch something very cold (heat flows out of us into the cold object). As heat flows from the hotter body to the colder body, we expect that the energy content and the temperatures of the two bodies will each change with time.

After some time being in thermal contact, we reach the situation in Fig. 4.1(c). The macroscopic properties of the two bodies are now no longer changing with time. If any energy flows from the first body to the second body, this is equal to the energy flowing from the second body to the first body; thus, there is *no net heat flow* between the two bodies. The two bodies are said to be in **thermal equilibrium**, which is defined by saying that the energy content and the temperatures of the

[1] This is assuming that no additional power is being fed into the systems, such as occurs in the operation of a refrigerator which sucks heat out of the cold interior and dumps it into your warmer kitchen, but only because you are supplying electrical power.

two bodies will no longer be changing with time. We would expect that the two bodies in thermal equilibrium are now at the *same temperature*.

It seems that something irreversible has happened. Once the two bodies were put in thermal contact, the change from Fig. 4.1(b) to Fig. 4.1(c) proceeds inevitably. However, if we started with two bodies at the same temperature and placed them in thermal contact as in Fig. 4.1(c), the reverse process, i.e. ending up with Fig. 4.1(b), would not occur.[2] Thus as a function of time, systems in thermal contact tend towards thermal equilibrium, rather than away from it. The process that leads to thermal equilibrium is called **thermalization**.

[2] Thermal processes thus define an arrow of time. We will return to this point later in Section 34.5.

If various bodies are all in thermal equilibrium with each other, then we would expect that their temperatures should be the same. This idea is encapsulated in the **zeroth law of thermodynamics**, which states that

> **Zeroth law of thermodynamics**:
> Two systems, each separately in thermal equilibrium with a third, are in equilibrium with each other.

You can tell by the numbering of the law that although it is an assumption that comes before the other laws of thermodynamics, it was added after the first three laws had been formulated. Early workers in thermodynamics took the content of the zeroth law as so obvious it hardly needed stating, and you might well agree with them! Nevertheless, the zeroth law gives us some justification for how to actually measure temperature: we place the body whose temperature needs to be measured in thermal contact with a second body which displays some property which has a well-known dependence on temperature and wait for them to come into thermal equilibrium. The second body is called a **thermometer**. The zeroth law then guarantees that if we have calibrated this second body against any other standard thermometer, we should always get consistent results. Thus, a more succinct statement of the zeroth law[3] is: 'thermometers work'.

[3] This version is from our colleague M.G. Bowler.

4.2 Thermometers

We now make some remarks concerning thermometers.

- For a thermometer to work well, its heat capacity must be much lower than that of the object whose temperature one wants to measure. If this is not the case, the action of measurement (placing the thermometer in thermal contact with the object) could alter the temperature of the object.
- A common type of thermometer utilizes the fact that liquids expand when they are heated. Galileo Galilei used a water thermometer based on this principle in 1593, but it was Daniel Gabriel Fahrenheit (1686–1736) who devised thermometers based on alco-

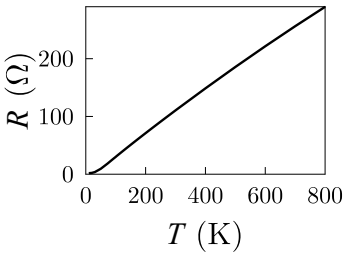


Fig. 4.2 The temperature dependence of the resistance of a typical platinum sensor.

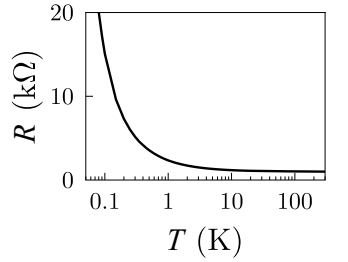


Fig. 4.3 The temperature dependence of the resistance of a typical RuO_2 sensor.

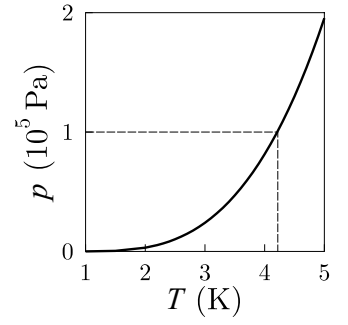


Fig. 4.4 The vapour pressure of ^{4}He as a function of temperature. The dashed line labels atmospheric pressure and the corresponding boiling point for liquid ^{4}He.

hol (1709) and mercury (1714) that bear most resemblance to modern household thermometers. He introduced his famous temperature scale which was then superseded by the more logical scheme devised by Anders Celsius (1701–1744).

- Another method is to measure the electrical resistance of a material which has a well-known dependence of resistance on temperature. Platinum is a popular choice since it is chemically resistant, ductile (so can be easily drawn into wires) and has a large temperature-coefficient of resistance; see Fig. 4.2. Other commonly used thermometers are based on doped germanium (a semiconductor which is very stable after repeated thermal cycling), carbon sensors and RuO_2 (in contrast to platinum, the electrical resistance of these thermometers increases as they are cooled; see Fig. 4.3).

- Using the ideal gas equation (eqn 1.12), one can measure the temperature of a gas by measuring its pressure with its volume fixed (or by measuring its volume with its pressure fixed). This works well as far as the ideal gas equation works, although at very low temperature, gases liquefy and show departures from the ideal gas equation.

- Another method which is useful in cryogenics is to have a liquid coexisting with its vapour and to measure the vapour pressure. For example, liquid helium (^{4}He, the most common isotope) has a vapour pressure dependence on temperature which is shown in Fig. 4.4.

All of these methods use some measurable property, like resistance or pressure, which depends in some, sometimes complicated, manner on temperature. However, none of them are completely linear across the entire temperature range of interest: mercury solidifies at very low temperature and becomes gaseous at very high temperature, the resistance of platinum saturates at very low temperature and platinum wire melts at very high temperature, etc. However, against what standard thermometer can one possibly assess the relative merits of these different thermometers? Which thermometer is perfect and gives the real thing, against which all other thermometers should be judged?

It is clear that we need some absolute definition of temperature based on fundamental physics. In the nineteenth century, one such definition was found, and it was based on a hypothetical machine, which has never been built, called a *Carnot engine*.[4] Subsequently, it was found that temperature could be defined in terms of a purely statistical argument using ideas from probability theory, and this is the one we will use which we introduce in Section 4.4. In the following section we will introduce the terminology of *microstates* and *macrostates* that will be needed for this argument.

[4]We will introduce the Carnot engine in Section 13.2. The definition of temperature which arises from this is based upon eqn 13.7 and states that the ratio of the temperature of a body to the heat flow from it is a constant in a reversible Carnot cycle.

4.3 The microstates and macrostates

To make the distinction between microstates and macrostates, consider the following example.

Example 4.1

Imagine that you have a large box containing 100 identical coins. With the lid on the box, you give it a really good long and hard shake, so that you can hear the coins flipping, rattling and being generally tossed around. Now you open the lid and look inside the box. Some of the coins will be lying with heads facing up and some with tails facing up. There are lots of possible configurations that one could achieve (2^{100} to be precise, which is approximately 10^{30}) and we will assume that *each of these different configurations is equally likely.* Each possible configuration therefore has a probability of approximately 10^{-30}. We will call each particular configuration a **microstate** of this system. An example of one of these microstates would be: 'Coin number 1 is heads, coin number 2 is heads, coin number 3 is tails, etc'. To identify a microstate, you would somehow need to identify each coin individually, which would be a bit of a bore. However, probably the way you would categorize the outcome of this experiment is by simply counting the number of coins which are heads and the number which are tails (e.g. 53 heads and 47 tails). This sort of categorisation we call a **macrostate** of this system. The macrostates are not equally likely. For example, of the $\approx 10^{30}$ possible individual configurations (microstates),

$$\begin{aligned}
&\#\text{ of configurations with 50 heads and 50 tails} &&= \tfrac{100!}{(50!)^2} \approx 4\times 10^{27},\\
&\#\text{ of configurations with 53 heads and 47 tails} &&= \tfrac{100!}{53!47!} \approx 3\times 10^{27},\\
&\#\text{ of configurations with 90 heads and 10 tails} &&= \tfrac{100!}{90!10!} \approx 10^{13},\\
&\#\text{ of configurations with 100 heads and 0 tails} &&= 1.
\end{aligned}$$

Thus, the outcome with all 100 coins with heads facing up is a very unlikely outcome. This macrostate contains a single microstate. If that were the result of the experiment, you would probably conclude that (i) your shaking had not been very vigorous and that (ii) someone had carefully prepared the coins to be lying heads up at the start of the experiment. Of course, a *particular* microstate with 53 heads and 47 tails is just as unlikely; it is just that there are about 3×10^{27} other microstates which have 53 heads and 47 tails which look extremely similar.

This simple example shows two crucial points:

- The system could be described by a very large number of *equally likely* microstates.
- What you actually measure[5] is a property of the macrostate of the

[5] In our example, the measurement was opening the large box and counting the number of coins which were heads and those which were tails.

system. The macrostates are *not* equally likely, because different macrostates correspond to different numbers of microstates.

The most likely macrostate that the system will find itself in is the one which corresponds to the largest number of microstates.

Thermal systems behave in a very similar way to the example we have just considered. To specify a microstate for a thermal system, you would need to give the microscopic configurations (perhaps position and velocity, or perhaps energy) of each and every atom in the system. In general it is impossible to measure which microstate the system is in. The macrostate of a thermal system on the other hand would be specified only by giving the macroscopic properties of the system, such as the pressure, the total energy or the volume. A macroscopic configuration, such as a gas with pressure $10^5\,\mathrm{Pa}$ in a volume $1\,\mathrm{m}^3$, would be associated with an enormous number of microstates. In the next section, we are going to give a statistical definition of temperature which is based on the idea that a thermal system can have a large number of equally likely microstates, but you are only able to measure the macrostate of the system. At this stage, we are *not* going to worry about what the microstates of the system actually are; we are simply going to posit their existence and say that if the system has energy E, then it could be in any one of $\Omega(E)$ equally likely microstates, where $\Omega(E)$ is some enormous number.

4.4 A statistical definition of temperature

We return to our example of Section 4.1 and consider two large systems which can exchange energy with each other, but not with anything else (Fig. 4.5). In other words, the two systems are in thermal contact with each other, but thermally isolated from their surroundings. The first system has energy E_1 and the second system has energy E_2. The total energy $E = E_1 + E_2$ is therefore assumed fixed since the two systems cannot exchange energy with anything else. Hence the value of E_1 is enough to determine the macrostate of this joint system. Each of these systems can be in a number of possible microstates. This number of possible microstates could in principle be calculated as in Section 1.4 (and in particular, Example 1.3) and will be a very large, combinatorial number, but we will not worry about the details of this. Let us assume that the first system can be in any one of $\Omega_1(E_1)$ microstates and the second system can be in any one of $\Omega_2(E_2)$ microstates. Thus the whole system can be in any one of $\Omega_1(E_1)\Omega_2(E_2)$ microstates.[6]

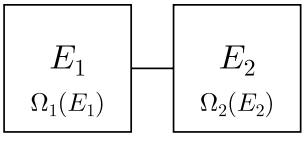


Fig. 4.5 Two systems able to exchange energy between them.

[6] We use the product of the two quantities, $\Omega_1(E_1)$ and $\Omega_2(E_2)$, because for each of the $\Omega_1(E_1)$ states of the first system, the second system can be in any of its $\Omega_2(E_2)$ different states. Hence the total number of possible combined states is the product of $\Omega_1(E_1)$ and $\Omega_2(E_2)$.

The systems are able to exchange energy with each other, and we will assume that they have been left in the condition of being joined together for a sufficiently long time that they have come into *thermal equilibrium*. This means that E_1 and E_2 have come to fixed values. The crucial insight which we must make is that *a system will appear to choose a macroscopic configuration which maximizes the number of microstates*. This idea is based upon the following assumptions:

(1) each one of the possible microstates of a system is equally likely to occur;
(2) the system's internal dynamics are such that the microstates of the system are continually changing;
(3) given enough time, the system will explore all possible microstates and spend an equal time in each of them.[7]

[7] This is the so-called ergodic hypothesis.

These assumptions imply that the system will most likely be found in a configuration which is represented by the most microstates. For a large system our phrase 'most likely' becomes 'absolutely, overwhelmingly likely'; what appears at first sight to be a somewhat weak, probabilistic statement (perhaps on the same level as a five-day weather forecast) becomes an utterly reliable prediction on whose basis you can design an aircraft engine and trust your life to it!

For our problem of two connected systems, the most probable division of energy between the two systems is the one which maximizes $\Omega_1(E_1)\Omega_2(E_2)$, because this will correspond to the greatest number of possible microstates. Our systems are large and hence we can use calculus to study their properties; we can therefore consider making infinitesimal changes to the energy of one of the systems and seeing what happens. Therefore, we can maximize this expression with respect to E_1 by writing

$$\frac{\mathrm{d}}{\mathrm{d}E_1}\left(\Omega_1(E_1)\Omega_2(E_2)\right) = 0 \tag{4.1}$$

and hence, using standard rules for differentiation of a product, we have

$$\Omega_2(E_2)\frac{\mathrm{d}\Omega_1(E_1)}{\mathrm{d}E_1} + \Omega_1(E_1)\frac{\mathrm{d}\Omega_2(E_2)}{\mathrm{d}E_2}\frac{\mathrm{d}E_2}{\mathrm{d}E_1} = 0. \tag{4.2}$$

Since the total energy $E = E_1 + E_2$ is assumed fixed, this implies that

$$\mathrm{d}E_1 = -\mathrm{d}E_2, \tag{4.3}$$

and hence

$$\frac{\mathrm{d}E_2}{\mathrm{d}E_1} = -1, \tag{4.4}$$

so that eqn 4.2 becomes

$$\frac{1}{\Omega_1}\frac{\mathrm{d}\Omega_1}{\mathrm{d}E_1} - \frac{1}{\Omega_2}\frac{\mathrm{d}\Omega_2}{\mathrm{d}E_2} = 0, \tag{4.5}$$

and hence

$$\frac{\mathrm{d}\ln\Omega_1}{\mathrm{d}E_1} = \frac{\mathrm{d}\ln\Omega_2}{\mathrm{d}E_2}. \tag{4.6}$$

This condition defines the most likely division of energy between the two systems if they are allowed to exchange energy since it maximizes the total number of microstates. This division of energy is, of course, more usually called 'being at the same temperature', and so we identify $\mathrm{d}\ln\Omega/\mathrm{d}E$ with the temperature T (so that our two systems have $T_1 = T_2$). We will define the **temperature** T by

$$\frac{1}{k_\mathrm{B}T} = \frac{\mathrm{d}\ln\Omega}{\mathrm{d}E}, \tag{4.7}$$

where k_B is the **Boltzmann constant**, which is given by

$$k_B = 1.3807 \times 10^{-23}\,\mathrm{J\,K^{-1}}. \tag{4.8}$$

With this choice of constant, T has its usual interpretation and is measured in Kelvin. We will show in later chapters that this choice of definition leads to experimentally verifiable consequences, such as the correct expression for the pressure of a gas.

We will see later (Section 14.5) that in statistical mechanics, the quantity $k_B \ln \Omega$ is called the **entropy**, S, and hence eqn 4.7 is equivalent to

$$\frac{1}{T} = \frac{\mathrm{d}S}{\mathrm{d}E}.$$

4.5 Ensembles

We are using probability to describe thermal systems and our approach is to imagine repeating an experiment to measure a property of a system again and again because we cannot control the microscopic properties (as described by the system's microstates). In an attempt to formalize this, Josiah Willard Gibbs in 1878 introduced a concept known as an **ensemble**. This is an idealization in which one consider making a large number of mental 'photocopies' of the system, each one of which represents a possible state the system could be in. There are three main ensembles that tend to be used in thermal physics:

(1) The **microcanonical ensemble**: an ensemble of systems that each have the same fixed energy.

(2) The **canonical ensemble**: an ensemble of systems, each of which can exchange its energy with a large reservoir of heat. As we shall see, this fixes (and defines) the temperature of the system.

(3) The **grand canonical ensemble**: an ensemble of systems, each of which can exchange both energy and particles with a large reservoir. (This fixes the system's temperature and a quantity known as the system's chemical potential. We will not consider this again until Chapter 22 and it can be ignored for the present.)

In the next section we will consider the canonical ensemble in more detail and use it to derive the probability of a system at a fixed temperature being in a particular microstate.

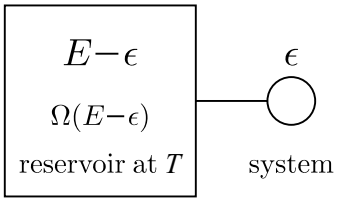


Fig. 4.6 A large reservoir (or heat bath) at temperature T connected to a small system.

4.6 Canonical ensemble

We now consider two systems coupled as before in such a way that they can exchange energy (Fig. 4.6). This time, we will make one of them enormous, and call it the **reservoir** (also known as a **heat bath**). It is so large that you can take quite a lot of energy out of it and yet it can remain at essentially the same temperature. In the same way, if you stand on the sea shore and take an eggcup-full of water out of the ocean, you do not notice the level of the ocean going down (although it does in fact go down, but by an unmeasurably small amount). The number of ways of arranging the quanta of energy of the reservoir will therefore be colossal. The other system is small and will be known as

the **system**. We will assume that for each allowed energy of the system there is only a single microstate, and therefore the system has a value of Ω equal to one. Once again, we fix[8] the total energy of the system plus reservoir to be E. The energy of the reservoir is taken to be $E-\epsilon$ while the energy of the system is taken to be ϵ. This situation of a system in thermal contact with a large reservoir is very important and is known as the **canonical ensemble**.[9]

The probability $P(\epsilon)$ that the system has energy ϵ is proportional to the number of microstates which are accessible to the reservoir multiplied by the number of microstates which are accessible to the system. This is therefore

$$P(\epsilon) \propto \Omega(E-\epsilon) \times 1. \tag{4.9}$$

Since we have an expression for temperature in terms of the logarithm of Ω (eqn 4.7), and since $\epsilon \ll E$, we can perform a Taylor expansion[10] of $\ln\Omega(E-\epsilon)$ around $\epsilon = 0$, so that

$$\ln\Omega(E-\epsilon) = \ln\Omega(E) - \frac{\mathrm{d}\ln\Omega(E)}{\mathrm{d}E}\epsilon + \cdots \tag{4.10}$$

and so now using eqn. 4.7, we have

$$\ln\Omega(E-\epsilon) = \ln\Omega(E) - \frac{\epsilon}{k_{\mathrm{B}}T} + \cdots, \tag{4.11}$$

where T is the temperature of the reservoir. In fact, we can neglect the further terms in the Taylor expansion (see Exercise 4.4) and hence eqn 4.11 becomes

$$\Omega(E-\epsilon) = \Omega(E)\,\mathrm{e}^{-\epsilon/k_{\mathrm{B}}T}. \tag{4.12}$$

Using eqn 4.9 we thus arrive at the following result for the probability distribution describing the system which is given by

$$P(\epsilon) \propto \mathrm{e}^{-\epsilon/k_{\mathrm{B}}T}. \tag{4.13}$$

Since the system is now in equilibrium with the reservoir, it also must have the same temperature as the reservoir. But notice that although the system therefore has fixed temperature T, its energy ϵ is not a constant but is governed by the probability distribution in eqn 4.13 (and is plotted in Fig. 4.7). This is known as the **Boltzmann distribution** and also as the **canonical distribution**. The term $\mathrm{e}^{-\epsilon/k_{\mathrm{B}}T}$ is known as a **Boltzmann factor**.

We now have a probability distribution which describes exactly how a small system behaves when coupled to a large reservoir at temperature T. The system has a reasonable chance of achieving an energy ϵ which is less than $k_{\mathrm{B}}T$, but the exponential in the Boltzmann distribution begins to quickly reduce the probability of achieving an energy much greater than $k_{\mathrm{B}}T$. However, to quantify this properly we need to normalize the probability distribution. If a system is in contact with a reservoir and has a microstate r with energy E_r, then

$$P(\text{microstate } r) = \frac{\mathrm{e}^{-E_r/k_{\mathrm{B}}T}}{\sum_i \mathrm{e}^{-E_i/k_{\mathrm{B}}T}}, \tag{4.14}$$

[8] We thus treat the system plus reservoir as being in what is known as the **microcanonical ensemble**, which has fixed energy with each of its microstates being equally likely.

[9] 'Canonical' means part of the 'canon', the store of generally accepted things one should know. It's an odd word, but we're stuck with it. Focussing on a system whose energy is *not* fixed, but which can exchange energy with a big reservoir, is something we do a lot in thermal physics and is therefore in some sense canonical.

[10] See Appendix B.

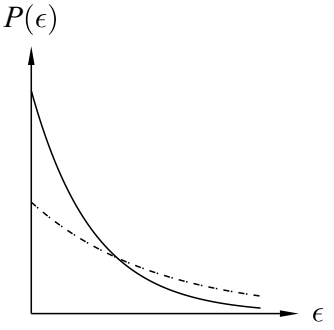


Fig. 4.7 The Boltzmann distribution. The dashed curve corresponds to a higher temperature than the solid curve.

where the sum in the denominator makes sure that the probability is normalized. The sum in the denominator is called the **partition function** and is given the symbol Z.

The partition function is the subject of Chapter 20.

We have derived the Boltzmann distribution on the basis of statistical arguments which show that this distribution of energy maximizes the number of microstates. It is instructive to verify this for a small system, so the following example presents the results of a computer experiment to demonstrate the validity of the Boltzmann distribution.

Example 4.2

To illustrate the statistical nature of the Boltzmann distribution, let us play a game in which quanta of energy are distributed in a lattice. We choose a lattice of 400 sites, arranged for convenience on a 20×20 grid. Each site initially contains a single energy quantum, as shown in Fig. 4.8(a). The adjacent histogram shows that there are 400 sites with one quantum on each. We now choose a site at random and remove the quantum from that site and place it on a second, randomly-chosen site. The resulting distribution is shown in Fig. 4.8(b), and the histogram shows that we now have 398 sites each with 1 quantum, 1 site with no quanta and 1 site with two quanta. This redistribution process is repeated many times and the resulting distribution is as shown in Fig. 4.8(c). The histogram describing this looks very much like a Boltzmann exponential distribution.

The initial distribution shown in Fig. 4.8(a) is very equitable and gives a distribution of energy quanta between sites of which Karl Marx would have been proud. It is however very *statistically unlikely* because it is associated with only a single microstate, i.e. $\Omega = 1$. There are many more microstates associated with other macrostates, as we shall now show. For example, the state obtained after a single iteration, such as the one shown in Fig. 4.8(b), is much more likely, since there are 400 ways to choose the site from which a quantum has been removed, and then 399 ways to choose the site to which a quantum is added; hence $\Omega = 400 \times 399 = 19600$ for this histogram (which contains 398 singly occupied sites, one site with zero quanta and one site with two quanta). The state obtained after many iterations in Fig. 4.8(c) is much, much more likely to occur if quanta are allowed to rearrange randomly as the number of microstates associated with the Boltzmann distribution is absolutely enormous. The Boltzmann distribution is simply a matter of probability.

In the model considered in this example, the rôle of temperature is played by the total number of energy quanta in play. So, for example, if instead the initial arrangement had been two quanta per site rather than one quantum per site, then after many iterations one would obtain the arrangement shown in Fig. 4.8(d). Since the initial arrangement has more energy, the final state is a Boltzmann distribution with a higher temperature (leading to more sites with more energy quanta).

11 Energy

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In this chapter we are going to focus on one of the key concepts in thermal physics, that of *energy*. What happens when energy is changed from one form to another? How much work can you get out of a quantity of heat? These are key questions to be answered. We are now beginning a study of **thermodynamics** proper, and in this chapter we will introduce the first law of thermodynamics. Before the first law, the most important concept in this chapter, we will introduce some additional ideas.

11.1 Some definitions

11.1.1 A system in thermal equilibrium

In thermodynamics, we define a **system** to be whatever part of the Universe we select for study. Near the system are its **surroundings**. We recall from Section 4.1 that a system is in *thermal equilibrium* when its macroscopic observables (such as its pressure or its temperature) have ceased to change with time. If you take a gas in a container which has been held at a certain stable temperature for a considerable period of time, the gas is likely to be in thermal equilibrium. A system in thermal equilibrium having a particular set of macroscopic observables is said to be in a particular **equilibrium state**. If however, you *suddenly* apply a lot of heat to one side of the box, then *initially at least*, the gas is likely to be in a **non-equilibrium state**.

11.1.2 Functions of state

A system is in an equilibrium state if macroscopic observable properties have fixed, definite values, independent of 'how they got there'. These properties are **functions of state** (sometimes called **variables of state**). A **function of state** is any physical quantity that has a well-defined value for each equilibrium state of the system. Thus, in thermal equilibrium these variables of state have no time dependence. Examples are volume, pressure, temperature and internal energy, and we will introduce a lot more in what follows. Examples of quantities which are not functions of state include the position of particle number 4325667, the total work done on a system and the total heat put into the system. Below, we will show in detail why work and heat are not functions of state. However, the point can be understood as follows: the

fact that your hands are warm or cold depends on their current temperature (a function of state), independently of how you got them to that temperature. For example, you can get to the same final thermodynamic state of having warm hands by different combinations of working and heating, e.g. you can end up with warm hands by rubbing them together (using the muscles in your arms to do work on them) or putting them in a toaster[1] (adding heat).

We now give a more mathematical treatment of what is meant by a function of state. Let the state of a system be described by parameters $\boldsymbol{x} = (x_1, x_2, \ldots)$ and let $f(\boldsymbol{x})$ be some function of state. [Note that this could be a very trivial function, such as $f(\boldsymbol{x}) = x_1$, since what we've called 'parameters' are themselves functions of state. But we want to allow for more complicated functions of state which might be combinations of these 'parameters'.] Then if the system parameters change from $\boldsymbol{x}_\mathrm{i}$ to $\boldsymbol{x}_\mathrm{f}$, the change in f is

$$\Delta f = \int_{\boldsymbol{x}_\mathrm{i}}^{\boldsymbol{x}_\mathrm{f}} \mathrm{d}f = f(\boldsymbol{x}_\mathrm{f}) - f(\boldsymbol{x}_\mathrm{i}). \tag{11.1}$$

This only depends on the endpoints $\boldsymbol{x}_\mathrm{i}$ and $\boldsymbol{x}_\mathrm{f}$. The quantity $\mathrm{d}f$ is an **exact differential** (see Appendix C.7) and functions of state have exact differentials. By contrast, a quantity which is represented by an **inexact differential** is not a function of state. The following example illustrates these kinds of differentials.

[1] NB don't try this at home.

Example 11.1

Let a system be described by two parameters, x and y. Let $f = xy$ so that

$$\mathrm{d}f = \mathrm{d}(xy) = y\,\mathrm{d}x + x\,\mathrm{d}y. \tag{11.2}$$

Then if (x, y) changes from $(0, 0)$ to $(1, 1)$, the change in f is given by

$$\Delta f = \int_{(0,0)}^{(1,1)} \mathrm{d}f = [xy]_{(0,0)}^{(1,1)} = (1 \times 1) - (0 \times 0) = 1. \tag{11.3}$$

This answer is independent of the exact path taken (it could be any of those shown in Fig. 11.1) because $\mathrm{d}f$ is an exact differential.

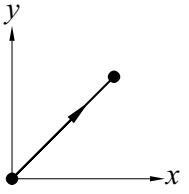
Now consider[2] $đg = y\,\mathrm{d}x$. The change in g when (x, y) changes from $(0, 0)$ to $(1, 1)$ along the path shown in Fig. 11.1(a) is given by

$$\Delta g = \int_{(0,0)}^{(1,1)} y\,\mathrm{d}x = \int_0^1 x\,\mathrm{d}x = \frac{1}{2}. \tag{11.4}$$

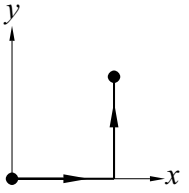
However when the integral is not carried out along the line $y = x$, but along the path shown in Fig. 11.1(b), it is given by

$$\Delta g = \int_{(0,0)}^{(1,0)} y\,\mathrm{d}x + \int_{(1,0)}^{(1,1)} y\,\mathrm{d}x = 0. \tag{11.5}$$

(a)



(b)



(c)

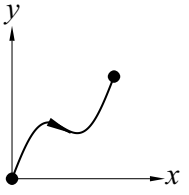


Fig. 11.1 Three possible paths between the points $(x, y) = (0, 0)$ and $(x, y) = (1, 1)$.

[2] We put a line through quantities such as the d in $đg$ to signify that it is an inexact differential.

If the integral is taken along the path shown in Fig. 11.1(c), yet another result would be obtained, but we are not going to attempt to calculate that!

Hence we find that the value of Δg depends on the path taken, and this is because đg is an inexact differential.[3]

[3]Note that if x is taken to be volume, V, and y is taken to be pressure, p, then the quantity f is proportional to temperature, while đg is the negative of the work đ$W = -p\,\mathrm{d}V$. This demonstrates that temperature is a function of state and work is not.

Recall from Section 1.2 that functions of state can either be:

- **extensive** (proportional to system size), e.g. energy, volume, magnetization, mass, or
- **intensive** (independent of system size), e.g. temperature, pressure, magnetic field, density, energy density.

In general one can find an **equation of state** which connects functions of state: for a gas this takes the form $f(p, V, T) = 0$. An example is the equation of state for an ideal gas, $pV = nRT$, which we met in eqn 1.12.

11.2 The first law of thermodynamics

Though the idea that heat and work are both forms of energy seems obvious to a modern physicist, the idea took some getting used to. Lavoisier had, in 1789, proposed that heat was a weightless, conserved fluid called *caloric*. Caloric was a fundamental element that couldn't be created or destroyed. Lavoisier's notion 'explained' a number of phenomena, such as combustion (fuels have stored caloric which is released on burning). Rumford in 1798 realised that something was wrong with the caloric theory: heating could be produced by friction, and if you keep on drilling through a cannon barrel (to take the example that drew the problem to his attention) almost limitless supplies of heat can be extracted. Where does all this caloric come from? Mayer quantified this in 1842 with an elegant experiment in which he frictionally generated heat in paper pulp and measured the temperature rise. Joule[4]independently performed similar experiments, but more accurately, in the period 1840–1845 (and his results became better known so that he was able to claim the credit!) Joule let a mass tied to a string slowly descend a certain height, while the other end of the string turns a paddle wheel immersed in a certain mass of water. The turning of the paddle frictionally heats the water. After a number of descents, Joule measured the temperature rise of the water. In this way he was able to deduce the 'mechanical equivalent of heat'. He also measured the heat output of a resistor (which, in modern units, is equal to I^2R, where I is the current and R the resistance). He was able to show that the same heat was produced for the same energy used, independent of the method of delivery. This implied that heat is a form of energy. Joule's experiments therefore consigned the caloric theory of heat to a footnote in history.

However, it was Mayer and later Helmholtz who elevated the experimental observations into a grand principle, which we can state as follows:

[4]The S.I. unit of energy is named after Joule. $1\,\mathrm{J} = 1\,\mathrm{N\,m}$. Older units are still in use in some places: 1 calorie is defined as the energy required to raise 1 g of water by 1°C (actually from 14.5°C to 15.5°C at sea level) and $1\,\mathrm{cal} = 4.184\,\mathrm{J}$. The energy contained in food is usually measured in kilocalories (kcal), where 1 kcal=1000 cal. Older books sometimes used the erg: $1\,\mathrm{erg} = 10^{-7}\,\mathrm{J}$. The British thermal unit (Btu) is an archaic unit, no longer commonly used in Britain: $1\,\mathrm{Btu} = 1055\,\mathrm{J}$. The foot-pound is $1\,\mathrm{ft\,lb} = 1.356\,\mathrm{J}$. Electricity bills often record energy in kilowatt hours ($1\,\mathrm{kWh} = 3.6\,\mathrm{MJ}$). Useful in atomic physics is the electron Volt: $1\,\mathrm{eV} = 1.602\times10^{-19}\,\mathrm{J}$.

The first law of thermodynamics:
Energy is conserved and heat and work are both forms of energy.

A system has an **internal energy** U which is the sum of the energy of all the internal degrees of freedom that the system possesses. U is a function of state because it has a well–defined value for each equilibrium state of the system. We can change the internal energy of the system by heating it or by doing work on it. The heat Q and work W are not functions of state since they concern the manner in which energy is delivered to (or extracted from) the system. After the event of delivering energy to the system, you have no way of telling which of Q or W was added to (or subtracted from) the system by examining the system's state.

The following analogy may be helpful: your personal bank balance behaves something like the internal energy U in that it acts like a function of state of your finances; cheques and cash are like heat and work in that they both result in a change in your bank balance, but after they have been paid in, you can't tell by simply looking at the value of your bank balance by which method the money was paid in.

The change in internal energy U of a system can be written

$$\Delta U = \Delta Q + \Delta W, \tag{11.6}$$

where ΔQ is the heat supplied **to** the system and ΔW is the work done **on** the system. Note the convention: ΔQ is positive for heat supplied to the system; if ΔQ is negative, heat is extracted from the system; ΔW is positive for work done on the system; if ΔW is negative, the system does work on its surroundings.

We define a **thermally isolated system** as a system which cannot exchange heat with its surroundings. In this case we find that $\Delta U = \Delta W$, because no heat can pass in or out of a thermally isolated system.

For a differential change, we write eqn 11.6 as

$$\boxed{\mathrm{d}U = đQ + đW,} \tag{11.7}$$

where $đW$ and $đQ$ are inexact differentials.

The work done on stretching a wire by a distance $\mathrm{d}x$ with a tension F is (see Fig. 11.2(a))

$$đW = F\,\mathrm{d}x. \tag{11.8}$$

The work done by compressing a gas (pressure p, volume V) by a piston can be calculated in a similar fashion (see Fig. 11.2(b)). In this case the force is $F = pA$, where A is the area of the piston, and $A\mathrm{d}x = -\mathrm{d}V$, so that

$$\boxed{đW = -p\,\mathrm{d}V.} \tag{11.9}$$

In this equation, the negative sign ensures that the work $đW$ done on the system is positive when $\mathrm{d}V$ is negative, i.e. when the gas is being compressed.

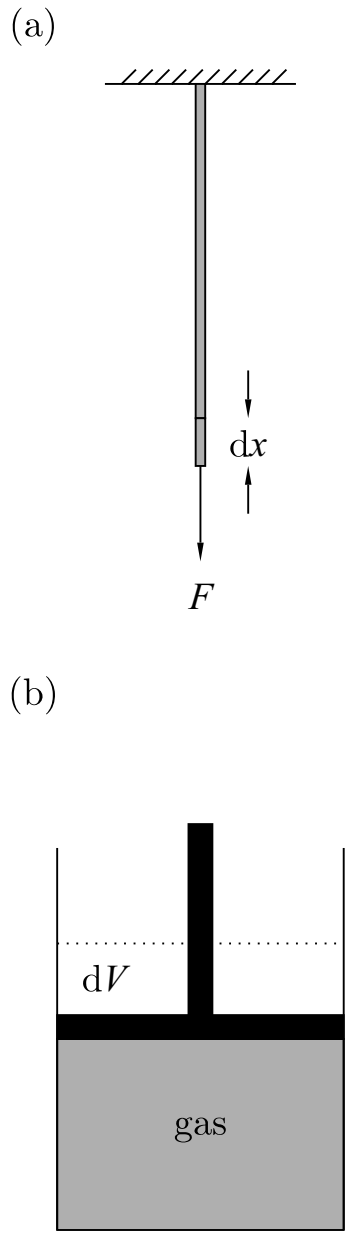


Fig. 11.2 (a) The work done stretching a wire by a distance $\mathrm{d}x$ is $F\,\mathrm{d}x$. (b) The work done compressing a gas is $-p\,\mathrm{d}V$.

It turns out that eqn 11.9 is only strictly true for a reversible change, a point we will explain further in Section 12.1. The idea is that if the piston is not frictionless, or if you move the piston too suddenly and generate shock waves, you will need to do more work to compress the gas because more heat is dissipated in the process.

11.3 Heat capacity

We now want to understand in greater detail how adding heat can change the internal energy of gas. In general, the internal energy will be a function of temperature and volume, so that we can write $U = U(T, V)$. Hence a small change in U can be related to changes in T and V by

$$\mathrm{d}U = \left(\frac{\partial U}{\partial T}\right)_V \mathrm{d}T + \left(\frac{\partial U}{\partial V}\right)_T \mathrm{d}V. \tag{11.10}$$

Rearranging eqn 11.7 with eqn 11.9 yields

$$đQ = \mathrm{d}U + p\,\mathrm{d}V, \tag{11.11}$$

and now using eqn 11.10 we have that

$$đQ = \left(\frac{\partial U}{\partial T}\right)_V \mathrm{d}T + \left[\left(\frac{\partial U}{\partial V}\right)_T + p\right] \mathrm{d}V. \tag{11.12}$$

We can divide eqn 11.12 by $\mathrm{d}T$ to obtain

$$\frac{đQ}{\mathrm{d}T} = \left(\frac{\partial U}{\partial T}\right)_V + \left[\left(\frac{\partial U}{\partial V}\right)_T + p\right] \frac{\mathrm{d}V}{\mathrm{d}T}, \tag{11.13}$$

which is valid for any change in T or V. However, what we want to know is what is the amount of heat we have to add to effect a change of temperature under certain constraints. The first constraint is that of keeping the volume constant. We recall the definition of the heat capacity at constant volume C_V (see Section 2.2, eqn 2.6) as

$$C_V = \left(\frac{\partial Q}{\partial T}\right)_V. \tag{11.14}$$

From eqn 11.13, this constraint knocks out the second term and implies that

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V. \tag{11.15}$$

The heat capacity at constant pressure is then, using eqns 2.7 and 11.13, given by

$$C_p = \left(\frac{\partial Q}{\partial T}\right)_p \tag{11.16}$$

$$= \left(\frac{\partial U}{\partial T}\right)_V + \left[\left(\frac{\partial U}{\partial V}\right)_T + p\right] \left(\frac{\partial V}{\partial T}\right)_p \tag{11.17}$$

so that

$$C_p - C_V = \left[\left(\frac{\partial U}{\partial V}\right)_T + p\right]\left(\frac{\partial V}{\partial T}\right)_p. \tag{11.18}$$

Recall from Section 2.2 that heat capacities are measured in $\mathrm{J\,K^{-1}}$ and refer to the heat capacity of a certain quantity of gas. We will sometimes wish to talk about the heat capacity per mole of gas, or sometimes the heat capacity per mass of gas. We will use small c for the latter, known as the *specific heat capacities*:

$$c_V = \frac{C_V}{M} \tag{11.19}$$

$$c_p = \frac{C_p}{M} \tag{11.20}$$

$$\tag{11.21}$$

where M is the mass of the material. Specific heat capacities are measured in $\mathrm{J\,K^{-1}\,kg^{-1}}$.

Example 11.2

Heat capacity of an ideal monatomic gas

For an ideal monatomic gas, the internal energy U is due to the kinetic energy, and hence $U = \frac{3}{2}RT$ per mole (see eqn 5.17; this result arises from the kinetic theory of gases). This means that U is only a function of temperature. Hence

$$\left(\frac{\partial U}{\partial V}\right)_T = 0. \tag{11.22}$$

The equation of state for 1 mole of ideal gas is

$$pV = RT, \tag{11.23}$$

so that

$$V = \frac{RT}{p}, \tag{11.24}$$

and hence

$$\left(\frac{\partial V}{\partial T}\right)_p = \frac{R}{p}, \tag{11.25}$$

and hence using eqns 11.18, 11.22 and 11.25 we have that

$$C_p - C_V = \left[\left(\frac{\partial U}{\partial V}\right)_T + p\right]\left(\frac{\partial V}{\partial T}\right)_p = R. \tag{11.26}$$

Because $U = \frac{3}{2}RT$, we therefore have that

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = \frac{3}{2}R \text{ per mole}, \tag{11.27}$$

and

$$C_p = C_V + R = \frac{5}{2}R \text{ per mole}. \tag{11.28}$$

Example 11.3

Is it always true that $\mathrm{d}U = C_V\,\mathrm{d}T$?
Solution: No, in general eqn 11.10 and eqn 11.15 imply that

$$\mathrm{d}U = C_V\,\mathrm{d}T + \left(\frac{\partial U}{\partial V}\right)_T \mathrm{d}V. \tag{11.29}$$

For an ideal gas, $\left(\frac{\partial U}{\partial V}\right)_T = 0$ (eqn 11.22) so it is true that

$$\mathrm{d}U = C_V\,\mathrm{d}T, \tag{11.30}$$

but for non-ideal gases, $\left(\frac{\partial U}{\partial V}\right)_T \neq 0$ and hence $\mathrm{d}U \neq C_v\,\mathrm{d}T$.

The ratio of C_p to C_V turns out to be a very useful quantity (we will see why in the following chapter) and therefore we give it a special name. Hence, we define the **adiabatic index**[5] γ as the *ratio* of C_p and C_V, so that

$$\gamma = \frac{C_p}{C_V}. \tag{11.31}$$

The reason for the name will become clear in the following chapter.

[5] γ is sometimes called the **adiabatic exponent**.

Example 11.4

What is γ for an ideal monatomic gas?
Solution: Using the results from the previous example[6]

$$\gamma = \frac{C_p}{C_V} = \frac{C_V + R}{C_V} = 1 + \frac{R}{C_V} = \frac{5}{3}. \tag{11.32}$$

[6] If the gas is not monatomic, γ can take a different value; see Section 19.2.

Example 11.5

Assuming $U = C_V T$ for an ideal gas, find (i) the internal energy per unit mass and (ii) the internal energy per unit volume.
Solution: Using the ideal gas equation $pV = Nk_\mathrm{B}T$ and the density $\rho = Nm/V$ (where m is the mass of one molecule), we find that

$$\frac{p}{\rho} = \frac{k_\mathrm{B}T}{m}. \tag{11.33}$$

Using eqn 11.32, we have that the heat capacity per mole is given by

$$C_V = \frac{R}{\gamma - 1}. \tag{11.34}$$

Hence, we can write that the internal energy for one mole of gas is

$$U = C_V T = \frac{RT}{\gamma - 1} = \frac{N_\mathrm{A} k_\mathrm{B} T}{\gamma - 1}. \tag{11.35}$$

The molar mass is mN_A, and so dividing eqn 11.35 by the molar mass, yields $\tilde{u}$, the internal energy per unit mass, given by

$$\tilde{u} = \frac{p}{\rho(\gamma - 1)}. \tag{11.36}$$

Multiplying $\tilde{u}$ by the density ρ gives u, the internal energy per unit volume, as

$$u = \rho\tilde{u} = \frac{p}{\gamma - 1}. \tag{11.37}$$

Chapter summary

- Functions of state have exact differentials.
- The first law of thermodynamics states that 'energy is conserved and heat is a form of energy'.
- $\mathrm{d}U = đW + đQ$.
- For a reversible change, $đW = -p\,\mathrm{d}V$.
- $C_V = \left(\frac{\partial Q}{\partial T}\right)_V = \left(\frac{\partial U}{\partial T}\right)_V$.
- $C_V = \left(\frac{\partial Q}{\partial T}\right)_P$ and $C_p - C_V = R$ for a mole of ideal gas.
- The adiabatic index is $\gamma = C_p/C_V$.

Exercises

(11.1) One mole of ideal monatomic gas is confined in a cylinder by a piston and is maintained at a constant temperature T_0 by thermal contact with a heat reservoir. The gas slowly expands from V_1 to V_2 while being held at the same temperature T_0. Why does the internal energy of the gas not change? Calculate the work done by the gas and the heat flow into the gas.

(11.2) Show that, for an ideal gas,

$$\frac{R}{C_V} = \gamma - 1 \tag{11.38}$$

and

$$\frac{R}{C_p} = \frac{\gamma - 1}{\gamma}, \tag{11.39}$$

where C_V and C_p are the heat capacities per mole.

(11.3) Consider the differential

$$\mathrm{d}z = 2xy\,\mathrm{d}x + (x^2 + 2y)\,\mathrm{d}y. \tag{11.40}$$

Evaluate the integral $\int_{(x_1,y_1)}^{(x_2,y_2)} \mathrm{d}z$ along the paths consisting of straight-line segments
(i) $(x_1, y_1) \to (x_2, y_1)$ and then $(x_2, y_1) \to (x_2, y_2)$.

(ii) $(x_1, y_1) \to (x_1, y_2)$ and then $(x_1, y_2) \to (x_2, y_2)$.
Is $\mathrm{d}z$ an exact differential?

(11.4) In polar coordinates, $x = r\cos\theta$ and $y = r\sin\theta$. The definition of x implies that

$$\frac{\partial x}{\partial r} = \cos\theta = \frac{x}{r}. \tag{11.41}$$

But we also have $x^2 + y^2 = r^2$, so differentiating with respect to r gives

$$2x\frac{\partial x}{\partial r} = 2r \qquad \Longrightarrow \qquad \frac{\partial x}{\partial r} = \frac{r}{x}. \tag{11.42}$$

But eqns 11.41 and 11.42 imply that

$$\frac{\partial x}{\partial r} = \frac{\partial r}{\partial x}. \tag{11.43}$$

What's gone wrong?

(11.5) In the comic song by Flanders and Swann about the laws of thermodynamics, they summarize the first law by the statement:

Heat is work and work is heat

Is that a good summary?

Antoine Lavoisier (1743–1794)

All flammable materials contain the odourless, colourless, tasteless substance *phlogiston*, and the process of burning them releases this phlogiston into the air. The burned material is said to be "dephlogistonated". That this notion is completely untrue was first shown by Antoine Lavoisier, who was born into a wealthy Parisian family. Lavoisier showed that both sulphur and phosphorous increase in weight once burned, but the weight gain was lost from the air. He demonstrated that it was oxygen which was responsible for combustion, not phlogiston and also that oxygen was responsible for the rusting of metals (his oxygen work was helped by results communicated to him by Joseph Priestley, and Lavoisier was a little lax in giving Priestley credit for this). Lavoisier showed that hydrogen and oxygen combined to make water and also identified the concept of an element as a fundamental substance that could not be broken down into simpler constituents by chemical processes. Lavoisier combined great experimental skill (and in this, he was ably assisted by his wife) and theoretical insight and is considered a founder of modern chemistry. Unfortunately, he added to his list of elemental substances both light and *caloric*, his proposed fluid which carried heat. Thus while ridding science of an unnecessary mythical substance (phlogiston), he introduced another one (caloric).



Fig. 11.3 Antoine Lavoisier

Lavoisier was a tax collector and thus found himself in the firing line when the French revolution started, the fact that he ploughed his dubiously-gotten gains into scientific research cutting no ice with revolutionaries. He had unfortunately made an enemy of Jean-Paul Marat, a journalist with an interest in science who in 1780 had wanted to join the French Academy of Sciences, but was blocked by Lavoisier. In 1792 Marat, now a firebrand revolutionary leader, demanded Lavoisier's death. Although Marat was himself assassinated in 1793 (while lying in his bath), Lavoisier was guillotined the following year.

Benjamin Thompson [Count Rumford] (1753–1814)

Thompson was born in rural Massachusetts and had an early interest in science. In 1772, as a humble doctor's apprentice he married a rich heiress, moved to Rumford, New Hampshire, and got himself appointed as a major in a local militia. He threw his lot in with the British during the American Revolution, feeding them information about the location of American forces and performing scientific work on the force of gunpowder. His British loyalties made him few friends in the land of his birth and he fled to Britain, abandoning his wife.



Fig. 11.4 Benjamin Thompson

He subsequently fell out with the British and moved, in 1785, to Bavaria where he worked for Elector Karl Theodor who made him a Count, and henceforth he was known as Count Rumford. He organised the poor workhouses, established the cultivation of the potato in Bavaria and invented Rumford soup. He continued to work on science, sometimes erratically (he believed that gases and liquids were perfect insulators of heat) but sometimes brilliantly; he noticed that the drilling of metal cannon barrels produced apparently limitless amounts of heat and his subsequent understanding of the production of heat by friction allowed him to put an end to Lavoisier's caloric theory. Not content with simply destroying Lavoisier's theory, he married Lavoisier's widow in 1804, though they separated four years later (Rumford unkindly remarked that Antoine Lavoisier had been lucky to have been guillotined than to have stayed married to her!). In 1799, Rumford founded the Royal Institution of Great Britain, establishing Davy as the first lecturer (Michael Faraday was appointed there 14 years later). He also endowed a medal for the Royal Society and a chair at Harvard. Rumford was also a prolific inventor and gave the world the Rumford fireplace, the double boiler, a drip coffeepot and, perhaps improbably, Baked Alaska (though Rumford's priority on the latter invention is not universally accepted).

12 Isothermal and adiabatic processes

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In this chapter we will apply the results of the previous chapter to illustrate some properties concerning isothermal and adiabatic expansions of gases. These results will assume that the expansions are reversible, and so the first part of this chapter explores the key concept of reversibility. This will be important for our discussion of entropy in subsequent chapters.

12.1 Reversibility

The laws of physics are reversible, so that if any process is allowed, then the time-reversed process can also occur. For example, if you could film the molecules in a gas bouncing off each other and the container walls, then when watching the film it would be hard to tell whether the film was being played forwards or backwards.

However, there are plenty of processes which you see in nature which seem to be irreversible. For example, consider an egg rolling off the edge of a table and smashing on the floor. Potential energy is converted into kinetic energy as the egg falls, and ultimately the energy ends up as a small amount of heat in the broken egg and the floor. The law of conservation of energy does not forbid the conversion of that heat back into kinetic energy of the reassembled egg which would then leap off the ground and back on to the table. However, this is never observed to happen. As another example, consider a battery which drives a current I through a resistor with resistance R and dissipates heat I^2R into the environment. Again, one never finds heat being absorbed by a resistor from its environment, resulting in the generation of a spontaneous current which can used to recharge the battery.

Lots of processes are like this, in which the final outcome is some potential, chemical or kinetic energy that gets converted into heat, which is then dissipated into the environment. As we shall see, the reason seems to be that there are lots more ways that the energy can be distributed in heat than in any other way, and this is therefore the most *probable outcome*. To try and understand this statistical nature of reversibility, it is helpful to consider the following example.

Example 12.1

We return to the situation described in Example 4.1. To recap, you are given a large box containing 100 identical coins. With the lid on the box, you give it a really good long and hard shake, so that you can hear the coins flipping, rattling and being generally tossed around. Now you open the lid and look inside the box. Some of the coins will be lying with heads facing up and some with tails facing up. We assume that each of the 2^{100} possible possible configurations (the microstates) are equally likely to be found. *Each of these is equally likely* and so each has a probability of occurrence of approximately 10^{-30}. However, the measurement made is counting the number of coins which are heads and the number which are tails (the macrostates), and the results of this measurement are not equally likely. In Example 3.1 we showed that of the $\approx 10^{30}$ individual microstates, a large number ($\approx 4 \times 10^{27}$) corresponded to 50 heads and 50 tails, but only one microstate corresponded to 100 heads and 0 tails.

Now, imagine that you had in fact carefully prepared the coins so that they were lying heads up. Following a good shake, the coins will most probably be a mixture of heads and tails. If, on the other hand, you carefully prepared a mixed arrangement of heads and tails, a good shake of the box is very unlikely to achieve a state in which all the coins lie with heads facing up. The process of shaking the box seems to *almost always* randomize the number of heads and tails, and this is an irreversible process.

This shows that the statistical behaviour of large systems is such as to make certain outcomes (such as a box of coins with mixed heads and tails) more likely than certain others (such as a box of coins containing coins the same way up). The statistics of large numbers therefore seems to drive many physical changes in an irreversible direction. How can we do a process in a reversible fashion?

The early researchers in thermodynamics wrestled with this problem which was of enormous practical importance in the design of engines in which you want to waste as little heat as possible in order to make your engine as efficient as possible. It was realized that when gases are expanded or compressed, it is possible to irreversibly convert energy into heat, and this will generally occur when we perform the expansion or the compression very fast, causing shock waves to be propagated through the gas (we will consider this effect in more detail in Chapter 32). However, it is possible to perform the expansion or compression *reversibly* if we do it sufficiently slowly so that *the gas remains in equilibrium throughout the entire process* and passes seamlessly from one equilibrium state to the next, each equilibrium state differing from the previous one by an infinitesimal change in the system parameters. Such a process is said to be **quasistatic**, since the process is almost in completely unchanging static equilibrium. As we shall see, heat can nevertheless be absorbed or

emitted in the process, while still maintaining reversibility.[1] In contrast, for an irreversible process, a non-zero change (rather than a sequence of infinitesimal changes) is made to the system, and therefore the system is not at equilibrium throughout the process.

An important (but given the name, perhaps not surprising) property of reversible processes is that you can run them in reverse. This fact we will use a great deal in Chapter 13. Of course, it would take an infinite amount of time for a strictly reversible process to occur, so most processes we term reversible are approximations to the 'real thing'.

[1] This is an important point: reversibility does not necessarily exclude the generation of heat. However, reversibility does require the absence of friction; a vehicle braking and coming to a complete stop, converting its kinetic energy into heat through friction in the brakes, is an irreversible process.

12.2 Isothermal expansion of an ideal gas

In this section, we will calculate the heat change in a reversible isothermal expansion of an ideal gas. The word **isothermal** means 'at constant temperature', and hence in an isothermal process

$$\Delta T = 0. \tag{12.1}$$

For an ideal gas, we showed in eqn 11.30 that $\mathrm{d}U = C_V\,\mathrm{d}T$, and so this means that for an isothermal change

$$\Delta U = 0, \tag{12.2}$$

since U is a function of temperature only. Equation 12.2 implies that $\mathrm{d}U = 0$ and hence from eqn 11.7

$$đW = -đQ, \tag{12.3}$$

so that the work done by the gas on its surroundings as it expands is equal to the heat absorbed by the gas. We can use $đW = -p\,\mathrm{d}V$ (eqn 11.9) which is the correct expression for the work done in a reversible expansion. Hence the heat absorbed by the gas during an isothermal expansion from volume V_1 to volume V_2 of 1 mole of an ideal gas at temperature T is

$$\begin{aligned}
\Delta Q &= \int đQ && (12.4)\\
&= -\int đW && (12.5)\\
&= \int_{V_1}^{V_2} p\,\mathrm{d}V && (12.6)\\
&= \int_{V_1}^{V_2} \frac{RT}{V}\,\mathrm{d}V && (12.7)\\
&= RT\ln\frac{V_2}{V_1}. && (12.8)
\end{aligned}$$

For an expansion, $V_2 > V_1$, and so $\Delta Q > 0$. The internal energy has stayed the same, but the volume has increased so that the energy density has gone down. The energy density and the pressure are proportional to one another[2], so that pressure will also have decreased.

[2] See eqn 6.25.

12.3 Adiabatic expansion of an ideal gas

The word **adiathermal** means 'without flow of heat'. A system bounded by adiathermal walls is said to be *thermally isolated*. Any work done on such a system produces an adiathermal change. We define a change to be **adiabatic** if it is both adiathermal and reversible. In an adiabatic expansion, therefore, there is no flow of heat and we have

$$đQ = 0. \tag{12.9}$$

The first law of thermodynamics therefore implies that

$$\mathrm{d}U = đW. \tag{12.10}$$

For an ideal gas, $\mathrm{d}U = C_V \mathrm{d}T$, and using $đW = -p\,\mathrm{d}V$ for a reversible change, we find that, for 1 mole of ideal gas,

$$C_V\,\mathrm{d}T = -p\,\mathrm{d}V = -\frac{RT}{V}\,\mathrm{d}V, \tag{12.11}$$

so that

$$\ln\frac{T_2}{T_1} = -\frac{R}{C_V}\ln\frac{V_2}{V_1}. \tag{12.12}$$

C_V here is per mole, since we are dealing with 1 mole of ideal gas.

Now $C_p = C_V + R$, and dividing this by C_V yields $\gamma = C_p/C_V = 1 + R/C_V$, and therefore $-(R/C_V) = 1 - \gamma$, so that eqn 12.12 becomes

$$TV^{\gamma-1} = \text{constant}, \tag{12.13}$$

or equivalently (using $pV \propto T$ for an ideal gas)

$$p^{1-\gamma}T^{\gamma} = \text{constant} \tag{12.14}$$

and

$$\boxed{pV^{\gamma} = \text{constant},} \tag{12.15}$$

the last equation probably being the most memorable.

Figure 12.1 shows **isotherms** (lines of constant temperature, as would be followed in an isothermal expansion) and **adiabats** (lines followed by an adiabatic expansion in which heat cannot enter or leave the system) for an ideal gas on a graph of pressure against volume. At each point, the adiabats have a steeper gradient than the isotherms, a fact we will return to in a later chapter.

12.4 Adiabatic atmosphere

The hydrostatic equation (eqn 4.23) expresses the additional pressure due to a thickness $\mathrm{d}z$ of atmosphere with density ρ and is

$$\mathrm{d}p = -\rho g\,\mathrm{d}z. \tag{12.16}$$

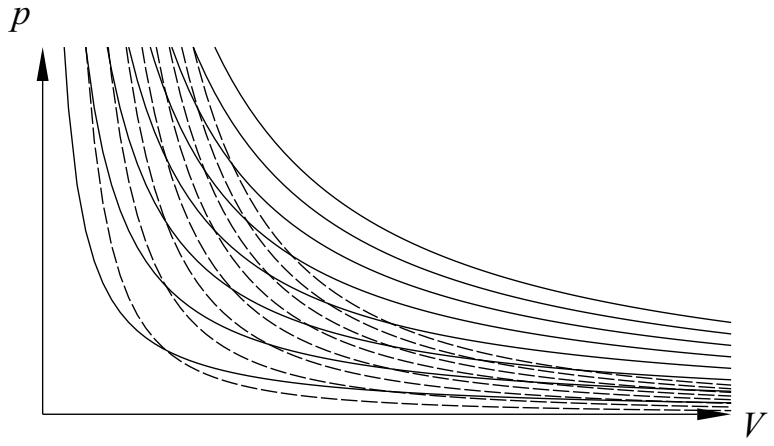


Fig. 12.1 Isotherms (solid lines) and adiabats (dashed lines).

Since $p = nk_{\mathrm{B}}T$ and $\rho = nm$, where m is the mass of one molecule, we can write $\rho = mp/k_{\mathrm{B}}T$ and hence

$$\frac{\mathrm{d}p}{\mathrm{d}z} = -\frac{mgp}{k_{\mathrm{B}}T}, \tag{12.17}$$

which implies that

$$T\frac{\mathrm{d}p}{p} = -\frac{mg}{k_{\mathrm{B}}}\mathrm{d}z. \tag{12.18}$$

For an isothermal atmosphere, T is a constant, and one obtains the results of Example 4.4. This assumes that the whole atmosphere is at a uniform temperature and is very unrealistic. A much better approximation (although nevertheless still an approximation to reality) is that each parcel of air[3] does not exchange heat with its surroundings. This means that if a parcel of air rises, it expands adiabatically. In this case, eqn 12.18 can be solved by recalling that for an adiabatic expansion $p^{1-\gamma}T^{\gamma}$ is a constant (see eqn 12.14) and hence that

$$(1-\gamma)\frac{\mathrm{d}p}{p} + \gamma\frac{\mathrm{d}T}{T} = 0. \tag{12.19}$$

Substituting this into eqn 12.18 yields

$$\frac{\mathrm{d}T}{\mathrm{d}z} = -\left(\frac{\gamma-1}{\gamma}\right)\frac{mg}{k_{\mathrm{B}}}, \tag{12.20}$$

which is an expression relating the rate of decrease of temperature with height, predicting it to be linear. We can rewrite $(\gamma-1)/\gamma = R/C_p$, and using $R = N_{\mathrm{A}}k_{\mathrm{B}}$ and writing the molar mass $M_{\mathrm{molar}} = N_{\mathrm{A}}m$ we can write eqn 12.20 as

$$\frac{\mathrm{d}T}{\mathrm{d}z} = -\frac{M_{\mathrm{molar}}g}{C_p}, \tag{12.21}$$

[3]Atmospheric physicists call a 'bit' of air a 'parcel'.

The quantity $M_{\text{molar}}g/C_p$ is known as the **adiabatic lapse rate**. For dry air (mostly nitrogen), it comes out as 9.7 K/km. Experimental values in the atmosphere are closer to 6–7 K/km (due partly to the fact that the atmosphere isn't dry, and latent heat effects, due to the heat needed to evaporate water droplets [and sometimes thaw ice crystals], are also important).

Chapter summary

- In an isothermal expansion $\Delta T = 0$.
- An adiabatic change is both adiathermal (no flow of heat) and reversible. In an adiabatic expansion of an ideal gas, pV^γ is constant.

Exercises

(12.1) In an adiabatic expansion of an ideal gas, pV^γ is constant. Show also that

$$TV^{\gamma-1} = \text{constant}, \tag{12.22}$$
$$T = \text{constant} \times p^{1-1/\gamma}. \tag{12.23}$$

(12.2) Assume that gases behave according to a law given by $pV = f(T)$, where $f(T)$ is a function of temperature. Show that this implies

$$\left(\frac{\partial p}{\partial T}\right)_V = \frac{1}{V}\frac{\mathrm{d}f}{\mathrm{d}T}, \tag{12.24}$$
$$\left(\frac{\partial V}{\partial T}\right)_p = \frac{1}{p}\frac{\mathrm{d}f}{\mathrm{d}T}. \tag{12.25}$$

Show also that

$$\left(\frac{\partial Q}{\partial V}\right)_p = C_p\left(\frac{\partial T}{\partial V}\right)_p, \tag{12.26}$$
$$\left(\frac{\partial Q}{\partial p}\right)_V = C_V\left(\frac{\partial T}{\partial p}\right)_V. \tag{12.27}$$

In an adiabatic change, we have that

$$\mathrm{d}Q = \left(\frac{\partial Q}{\partial p}\right)_V \mathrm{d}p + \left(\frac{\partial Q}{\partial V}\right)_p \mathrm{d}V = 0. \tag{12.28}$$

Hence show that pV^γ is a constant.

(12.3) Explain why we can write

$$đQ = C_p\,\mathrm{d}T + A\,\mathrm{d}p \quad \text{and} \tag{12.29}$$
$$đQ = C_V\,\mathrm{d}T + B\,\mathrm{d}V, \tag{12.30}$$

where A and B are constants. Subtract these equations and show that

$$(C_p - C_V)\mathrm{d}T = B\,\mathrm{d}V - A\,\mathrm{d}p, \tag{12.31}$$

and that at constant temperature

$$\left(\frac{\partial p}{\partial V}\right)_T = \frac{B}{A}. \tag{12.32}$$

In an adiabatic change, show that

$$\mathrm{d}p = -(C_p/A)\mathrm{d}T, \tag{12.33}$$
$$\mathrm{d}V = -(C_V/B)\mathrm{d}T. \tag{12.34}$$

Hence show that in an adiabatic change, we have that

$$\left(\frac{\partial p}{\partial V}\right)_{\text{adiabatic}} = \gamma\left(\frac{\partial p}{\partial V}\right)_T, \tag{12.35}$$
$$\left(\frac{\partial V}{\partial T}\right)_{\text{adiabatic}} = \frac{1}{1-\gamma}\left(\frac{\partial V}{\partial T}\right)_p, \tag{12.36}$$
$$\left(\frac{\partial p}{\partial T}\right)_{\text{adiabatic}} = \frac{\gamma}{\gamma-1}\left(\frac{\partial p}{\partial T}\right)_V \tag{12.37}$$

(12.4) Using eqn 12.35, relate the gradients of adiabats and isotherms on a p–V diagram.

(12.5) Two thermally insulated cylinders, A and B, of equal volume, both equipped with pistons, are connected by a valve. Initially A has its piston fully withdrawn and contains a perfect monatomic gas at temperature T, while B has its piston fully inserted, and the valve is closed. Calculate the final temperature of the gas after the following operations, which each start with the same initial arrangement. The thermal capacity of the cylinders is to be ignored.
(a) The valve is fully opened and the gas slowly drawn into B by pulling out the piston B; piston A remains stationary.
(b) Piston B is fully withdrawn and the valve is opened slightly; the gas is then driven as far as it will go into B by pushing home piston A at such a rate that the pressure in A remains constant: the cylinders are in thermal contact.

(12.6) In Rüchhardt's method of measuring γ, illustrated in Fig. 12.2, a ball of mass m is placed snugly inside a tube (cross-sectional area A) connected to a container of gas (volume V). The pressure p of the gas inside the container is slightly greater than atmospheric pressure p_0 because of the downward force of the ball, so that

$$p = p_0 + \frac{mg}{A}. \tag{12.38}$$

Show that if the ball is given a slight downwards displacement, it will undergo simple harmonic motion with period τ given by

$$\tau = 2\pi\sqrt{\frac{mV}{\gamma p A^2}}. \tag{12.39}$$

[You may neglect friction. As the oscillations are fairly rapid, the changes in p and V which occur can be treated as occurring adiabatically.]
In Rinkel's 1929 modification of this experiment, the ball is held in position in the neck where the gas pressure p in the container is exactly equal to air pressure, and then let drop, the distance L that it falls before it starts to go up again is measured. Show that this distance is given by

$$mgL = \frac{\gamma P A^2 L^2}{2V}. \tag{12.40}$$

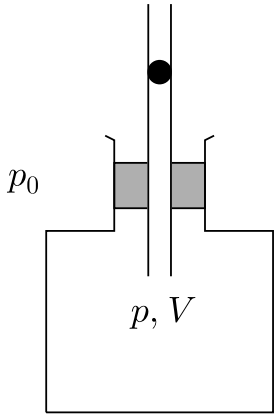


Fig. 12.2 Rüchhardt's apparatus for measuring γ. A ball of mass m oscillates up and down inside a tube.

Part V

The second law

In this part we introduce the *second law of thermodynamics* and follow its consequences. This part is structured as follows:

- In Chapter 13, we consider *heat engines*, which are cyclic processes that convert heat into work. We state various forms of the *second law of thermodynamics* and prove their equivalence, in particular showing that no engine can be more efficient than a *Carnot engine*. We also prove *Clausius' theorem*, which applies to any cyclic process.
- In Chapter 14 we show how the results from the preceding chapter lead to the concept of *entropy*. We derive the important equation $\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$, which combines the first and second laws of thermodynamics. We also introduce the *Joule expansion* and use it to discuss the statistical interpretation of entropy and *Maxwell's demon*.
- There is a very deep connection between *entropy* and *information*, and we explore this in Chapter 15, briefly touching upon *data compression* and *quantum information*.